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GEOSTATIONARY BENT-PIPE SATELLITE LINK BUDGET ANALYSIS & ATMOSPHERIC PROPAGATION STUDY

UZB451E – SPACECRAFT COMMUNICATIONS
TERM PROJECT REPORT

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1 Introduction

Satellite communication via geostationary orbit has been the backbone of intercontinental broadcasting, maritime connectivity, and VSAT business networks for over five decades. This term project presents an in-depth academic study of a modern Ku-band satellite link, mapping out the full physical layer from transmitter to receiver and quantifying the operational margins under realistic atmospheric conditions.

1.1 System Architecture and Ku-Band Principles

Placing a satellite in a circular equatorial orbit at an altitude of approximately 35,786 km ensures its orbital period matches the Earth's sidereal rotation. To a ground observer the satellite appears fixed at a single longitude, enabling inexpensive, non-tracking parabolic dish antennas. The price of this convenience is the immense free-space path loss (FSPL)—often exceeding 200 dB—accumulated over the $\approx$ 37,000 km slant range. Slight gravitational perturbations from the Sun and Moon also cause periodic station-keeping drifts that must be included in the system margin analysis.

The satellite payload modeled in this project is a transparent, classic *bent-pipe* transponder. Unlike regenerative payloads that demodulate and remodulate on-board, a bent-pipe satellite acts as an analog relay: it captures the weak uplink signal, filters, frequency-translates, amplifies via a TWTA or SSPA, and retransmits it.

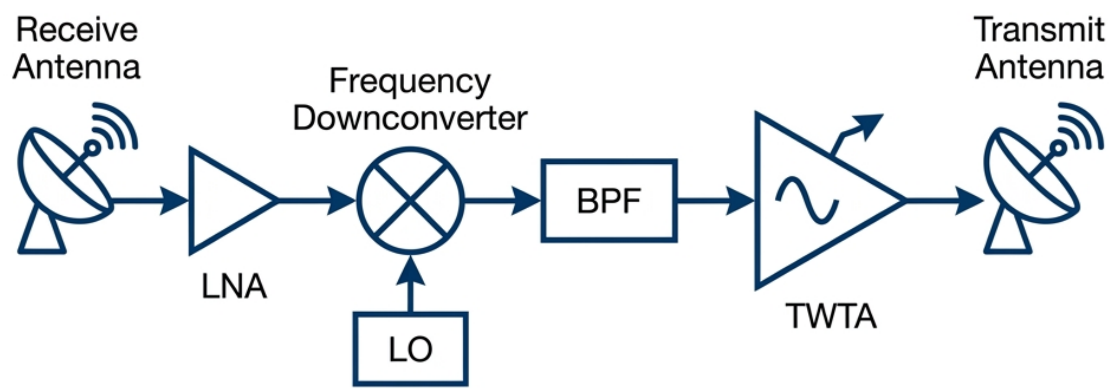


Figure 1: Functional block diagram of a classic transparent bent-pipe satellite transponder payload architecture.

This simplicity makes the satellite robust and format-agnostic but introduces a key challenge: uplink noise is amplified along with the desired signal, physically coupling the uplink and downlink budgets. The system engineer must ensure the combined end-to-end $(C/N_0)_{\text{total}}$ is sufficient for reliable demodulation.

To achieve this, the project uses a Ku-band link (uplink 14.0 GHz, downlink 12.0 GHz) with circular polarisation. The polarisation tilt angle $\tau = 45°$ is used in the ITU-R P.838-3 rain specific attenuation calculation as the standard averaging convention for circular polarisation; it does not imply a physical polarisation tilt. Ku-band is the commercial workhorse for DTH television and VSAT services, offering high antenna gain from compact dishes. Its main vulnerability is atmospheric hydrometeors: the signal wavelength (≈ 2 cm) is comparable to raindrop size, causing severe Mie scattering. This attenuation is modelled in Chapter 4 using the ITU-R propagation suite.

1.2 Scenario Definition

To ground the study in a realistic commercial context, we model a one-way transparent link between Rome and Ankara via the commercial **Eutelsat Hotbird 13G** satellite:

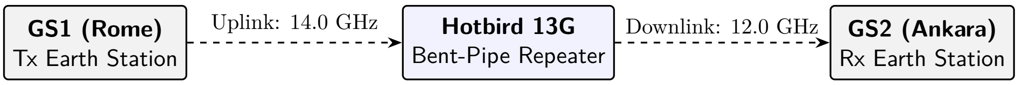


Figure 2: Scientific illustration of the Eutelsat Hotbird 13G telecommunication satellite in geostationary orbit, acting as our transparent bent-pipe orbital repeater.

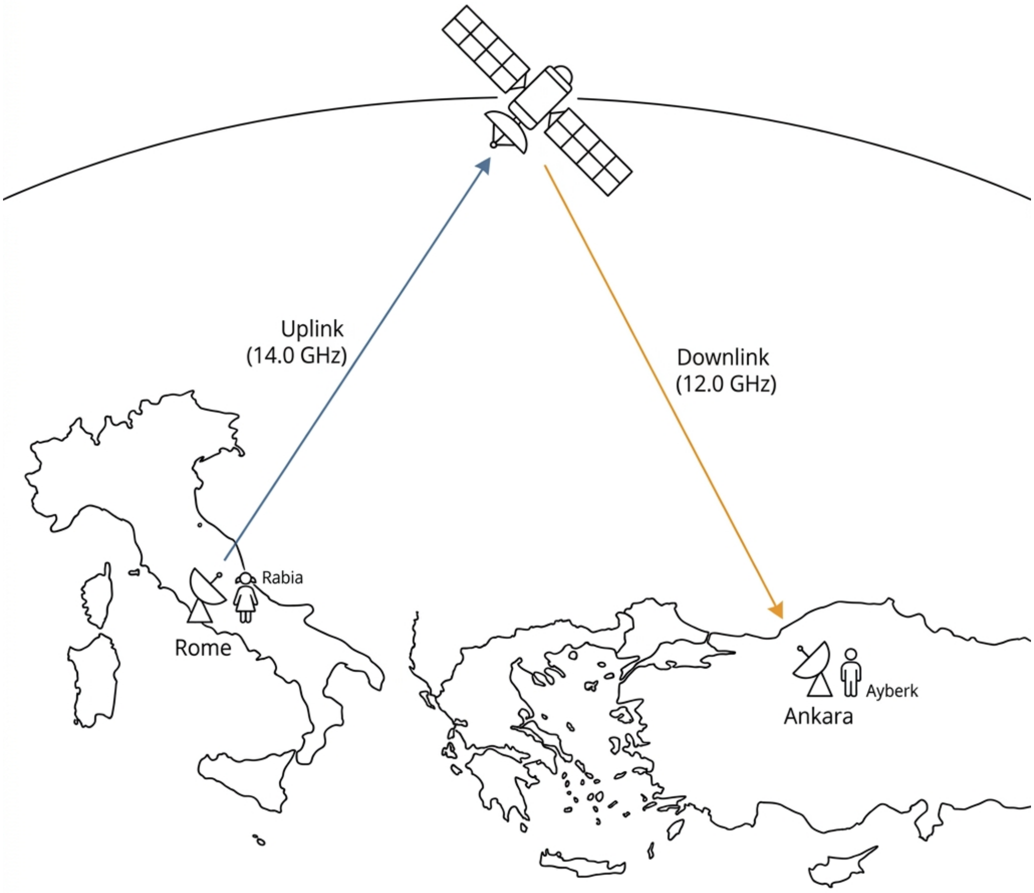


Figure 3: Architecture of the geostationary bent-pipe satellite link connecting GS1 Rome and GS2 Ankara via Eutelsat Hotbird 13G.

- **Ground Station 1 (GS1 Rome):** Located in Rome, Italy (41.9028° N, 12.4964° E). It acts as the broadcasting terminal (uplink station), utilizing a 2.4-meter parabolic dish antenna (datasheet-supported aperture [1]) with a 20.0 Watt high-power amplifier (HPA) assumed to represent a commercial Ku-band HPA at this power class.
- **GEO Satellite (Hotbird 13G):** Positioned at 13.0° E longitude on the geostationary equatorial arc. The satellite downlink EIRP at Ankara is estimated as 46 dBW from the HOTBIRD published footprint contour [2, 3]. The wider contour range is approximately 46–53 dBW; 46 dBW is used as a conservative baseline for the Ankara receiving site. Since public HOTBIRD documentation does not provide the selected transponder uplink receive G/T, a representative satellite receive G/T of 3.0 dB/K is adopted as an engineering assumption for the uplink budget.
- **Ground Station 2 (GS2 Ankara):** Located in Ankara, Turkey (39.9334° N, 32.8597° E). It acts as the receive-only downlink terminal, utilizing a 0.9-meter parabolic dish (datasheet-supported aperture [4]) with an assumed consumer-grade system noise temperature (T_{sys}) of 150 K. This value represents a typical outdoor

unit (LNB + feed) performance for a Ku-band consumer terminal; no public Tsys measurement for this specific LNB/antenna combination is available.

1.3 Computational Methodology

Although the original project statement mentioned a MATLAB Satellite Link Budget Analyzer as a reference workflow, the instructor explicitly permitted implementation in any programming language or computational environment. Therefore, this project implements the complete static and extended link-budget workflow in a custom Python analysis suite. The baseline equations follow the Pratt link design procedure [5], while the advanced layers extend the analysis with ITU-R atmospheric models (P.618, P.837, P.838, P.676, P.840), Monte Carlo rain-outage simulation, adjacent-satellite interference, dynamic system noise temperature, apparent GEO motion, and DVB-S2 ACM. This choice allows reproducible, parametric sensitivity studies and automated generation of contour plots directly from computed physical quantities.

1.4 Report Structure

Following this methodology, the report provides a structured, comprehensive analysis of the geostationary link. Chapter 2 details the classical link budget equations based on Pratt's textbook to establish our open-sky baseline. Chapter 3 presents advanced optional modeling layers (such as adjacent satellite interference, intermodulation distortion, digital BER/FEC, dynamic system noise temperature, and apparent GEO orbital drift). Chapter 4 models the standards-based ITU-R P.618 atmospheric suite alongside a DVB-S2 Adaptive Coding and Modulation (ACM) selector. Chapter 5 discusses the numerical results and provides detailed physical interpretations of our generated 2D contour maps, followed by a concluding summary in Chapter 6.

2 Link Design Equations

This chapter details the fundamental RF engineering equations and geometric transformations used to design the clear-sky static link budget. These models are derived based on the link design procedure outlined in *Chapter 4* of the reference textbook by Pratt, Bostian, and Allnutt [5]. The equations form the mathematical engine of our custom Python simulation suite.

2.1 Slant-Path Geometry and Coordinate Transformations

Calculating the pointing parameters (slant range, elevation angle, and azimuth angle) between a terrestrial ground station and a geostationary satellite requires robust transformations between spherical Earth coordinates and the Earth-Centered, Earth-Fixed (ECEF) Cartesian coordinate system.

Before locating the satellite, its nominal radius must be defined. According to Kepler's Third Law of Planetary Motion, the orbital radius r_{sat} for a geosynchronous orbit with period T (one sidereal day, ≈ 86164 seconds) is given by:

$$r_{sat} = \sqrt[3]{\frac{\mu \cdot T^2}{4\pi^2}} \approx 42164.0 \text{ km} \tag{1}$$

where $\mu \approx 3.986 \times 10^5$ km^3/s^2 is the Earth's standard gravitational parameter. Given the Earth's equatorial radius $R_e = 6378.137$ km, the satellite's altitude above the equator is approximately 35786 km.

For a ground station located at latitude ϕ, longitude λ, and altitude h above mean sea level, the ECEF position vector $\vec{R}_{ground}$ is:

$$\vec{R}_{ground} = \begin{bmatrix} (R_e + h)\cos\phi\cos\lambda \\ (R_e + h)\cos\phi\sin\lambda \\ (R_e + h)\sin\phi \end{bmatrix} \tag{2}$$

The geostationary satellite, stationed perfectly in the equatorial plane ($\phi_{sat} = 0°$) at longitude λ_{sat}, has the Cartesian position vector:

$$\vec{R}_{sat} = \begin{bmatrix} r_{sat}\cos\lambda_{sat} \\ r_{sat}\sin\lambda_{sat} \\ 0 \end{bmatrix} \tag{3}$$

The slant-range vector $\vec{\rho}$ pointing directly from the ground station to the satellite is computed via vector subtraction:

$$\vec{\rho} = \vec{R}_{sat} - \vec{R}_{ground} \tag{4}$$

The absolute physical distance (slant range) d, representing the electromagnetic wave's travel path, is the Euclidean norm of this vector:

$$d = \|\vec{\rho}\| = \sqrt{\rho_x^2 + \rho_y^2 + \rho_z^2} \tag{5}$$

It is worth noting that while this spherical Earth model offers exceptional simplicity, a highly precise geodetic-to-ECEF transformation would employ the WGS-84 ellipsoid parameters:

$$x = (N(\phi) + h)\cos\phi\cos\lambda \tag{6}$$
$$y = (N(\phi) + h)\cos\phi\sin\lambda \tag{7}$$
$$z = (N(\phi)(1 - e^2) + h)\sin\phi \tag{8}$$

where $e^2 = 2f - f^2$ is the first eccentricity squared ($f \approx 1/298.257$ is the flattening), and $N(\phi) = R_e/\sqrt{1 - e^2\sin^2\phi}$ is the prime vertical radius of curvature. However, for a satellite at geostationary altitude, the spherical approximation with $R_e \approx 6378.137$ km introduces a slant-range error of less than 0.1%, which is well within our system design margins.

To determine the pointing elevation angle θ, we project the unit line-of-sight vector $\vec{a}_\rho = \vec{\rho}/d$ onto the local topocentric unit vector pointing vertically Up ($\vec{u}_{up}$). The unit Up vector at the ground station coordinates is:

$$\vec{u}_{up} = \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix} \tag{9}$$

The local elevation angle θ, critical for calculating atmospheric attenuation path lengths, is then calculated as:

$$\theta = \arcsin(\vec{a}_\rho \cdot \vec{u}_{up}) = \arcsin\left(\frac{\vec{\rho} \cdot \vec{u}_{up}}{d}\right) \tag{10}$$

The azimuth angle Az is determined by projecting the line-of-sight vector onto the local East ($\vec{u}_{east}$) and North ($\vec{u}_{north}$) unit vectors, yielding the azimuth angle relative to true North:

$$Az = \arctan_2(\vec{a}_\rho \cdot \vec{u}_{east},\ \vec{a}_\rho \cdot \vec{u}_{north}) \tag{11}$$

2.2 Antenna Parameters and Aperture Efficiency

For a circular parabolic dish antenna of physical diameter D operating at carrier frequency f, the wavelength of the electromagnetic wave is $\lambda_w = c/f$, where c is the speed of light in vacuum. The peak boresight gain G in dBi is calculated as:

$$G(\text{dBi}) = 10\log_{10}\left(\eta\left(\frac{\pi D}{\lambda_w}\right)^2\right) \tag{12}$$

The parameter η represents the overall aperture efficiency. While an ideal antenna has an efficiency of 1.0 (100%), physical parabolic reflectors suffer from multiple combined electromagnetic losses:

$$\eta = \eta_{sub} \cdot \eta_{spill} \cdot \eta_{taper} \cdot \eta_{surface} \tag{13}$$

where η_{sub} represents sub-reflector blockage, η_{spill} is the spillover efficiency (fraction of feed power intercepted by the dish), η_{taper} is the amplitude illumination taper, and $\eta_{surface}$ is the surface roughness tolerance efficiency modeled by Ruze's rule. For our commercial baseline, a standard engineering efficiency of $\eta = 0.62$ (62%) is utilized.

The Half-Power Beamwidth (θ_{HPBW}), which dictates how tightly the antenna focuses its energy and dictates adjacent satellite interference susceptibility, is approximated by:

$$\theta_{HPBW} \approx 70^\circ \frac{\lambda_w}{D} \tag{14}$$

2.3 Free-Space Path Loss (FSPL) and Signal Spreading

The Free-Space Path Loss (FSPL) is often misunderstood as a resistive "loss" of energy to the vacuum. In reality, it is a purely geometric phenomenon resulting from the spherical spreading of the wavefront over the vast distance d. As the wave expands, its power density (Watts per square meter) drops proportionally to the surface area of the expanding sphere ($4\pi d^2$).

Derived directly from the Friis Transmission Equation, the free-space path loss L_{FSPL} in decibel terms is expressed as:

$$L_{\text{FSPL}}(\text{dB}) = 20\log_{10}\left(\frac{4\pi d}{\lambda_w}\right) = 92.45 + 20\log_{10}(f_{\text{GHz}}) + 20\log_{10}(d_{\text{km}}) \tag{15}$$

For a geostationary link, this term overwhelmingly dominates the power budget, typically ranging from 205 to 210 dB.

2.4 Additional System Losses: Pointing Error and Polarization Mismatch

While Free-Space Path Loss dominates the attenuation budget, real-world operational links suffer from additional impairments that must be strictly accounted for in L_{total}.

Antenna Pointing Error Loss (L_{point}): Even with precise mechanical installation, wind loading, thermal expansion, or satellite orbital drift can cause the ground antenna boresight to misalign with the satellite. For a small angular pointing error θ_{error} relative to the half-power beamwidth (θ_{HPBW}), the attenuation can be approximated using a Gaussian beam envelope:

$$L_{point}(\text{dB}) \approx 12\left(\frac{\theta_{error}}{\theta_{HPBW}}\right)^2 \tag{16}$$

Polarization Mismatch Loss (L_{pol}): When the polarization of the incoming wave does not perfectly align with the receiving antenna feed, signal power is lost. For circular polarization, imperfections in the antenna structure lead to a finite Cross-Polarization Discrimination (XPD). A typical well-designed reflector maintains an XPD of about 30 dB, representing the isolation between Right-Hand (RHCP) and Left-Hand (LHCP) states. Atmospheric depolarization, particularly during heavy rain (canting of raindrops), can further degrade XPD, introducing co-channel interference in frequency-reuse systems.

2.5 RF Link Power Budget and Receiver Figure of Merit

For any link segment, the Equivalent Isotropically Radiated Power (EIRP) defines the total effective power radiated in the direction of the main beam. It effectively multiplies the raw amplifier power by the focusing gain of the antenna:

$$EIRP(\text{dBW}) = P_{tx}(\text{dBW}) + G_{tx}(\text{dBi}) - L_{tx\_feeder}(\text{dB}) \tag{17}$$

At the receiving end, a useful receiver performance metric is the Figure of Merit (G/T), measured in dB/K. This parameter combines antenna gain and receiver noise temperature into a single figure of merit:

$$G/T(\text{dB/K}) = G_{\text{rx}}(\text{dBi}) - L_{\text{rx,feeder}}(\text{dB}) - 10\log_{10}(T_{\text{sys}}) \tag{18}$$

where T_{sys} is the comprehensive system noise temperature in Kelvin. The system noise temperature referred to the LNA input is modeled as:

$$T_{\text{sys}} = T_{\text{ant}} + T_{\text{feeder}} + (F_{\text{LNA}} - 1)T_0 \tag{19}$$

where T_{ant} is the antenna noise temperature (comprising cosmic sky noise and ground noise), T_{feeder} is the physical feeder waveguide loss noise, F_{LNA} is the noise factor of the Low-Noise Block amplifier, and $T_0 = 290$ K is the standard reference physical temperature. All noise contributions are referred to the LNA input port, which serves as the common noise reference plane throughout this analysis. The GS2 system noise temperature of 150 K is an input assumption representing the combined noise of a consumer-grade LNB at the Ankara terminal; the component breakdown (T_{ant}, T_{feeder}, T_{LNA}) is not independently specified as manufacturer noise data is not publicly available for this terminal class.

The fundamental Carrier-to-Noise Density ratio C/N_0 at the receiver demodulator input is synthesized using the link budget equation:

$$C/N_0(\text{dB-Hz}) = EIRP(\text{dBW}) + G/T(\text{dB/K}) - L_{total}(\text{dB}) - 10\log_{10}(k) \tag{20}$$

where L_{total} encompasses FSPL, pointing misalignments, polarization losses, and atmospheric clear-sky absorption. The term $-10\log_{10}(k) \approx 228.6$ dB represents Boltzmann's constant in decibels.

Ultimately, the Energy-per-Bit to Noise-Density ratio E_b/N_0 acts as the core digital indicator of link viability:

$$E_b/N_0(\text{dB}) = C/N_0(\text{dB-Hz}) - 10\log_{10}(R_b) \tag{21}$$

where R_b is the information transmission bit rate in bits per second (bps). The overall Link Margin, representing the system's safety cushion against fading events, is defined as:

$$Margin(\text{dB}) = E_b/N_0(\text{dB}) - \textit{Required}\ E_b/N_0(\text{dB}) \tag{22}$$

2.6 Uplink-Downlink Coupling in Bent-Pipe Transponders

Because a transparent geostationary bent-pipe transponder acts merely as an analog frequency-translating amplifier, it lacks the ability to distinguish between signal and noise. Any thermal noise, interference, or distortion captured by the satellite's receiving antenna on the uplink is indiscriminately amplified and retransmitted on the downlink.

Consequently, the end-to-end total Carrier-to-Noise Density ratio $(C/N_0)_{total}$ is mathematically coupled. It must be calculated via the inverse-linear addition of the individual uplink and downlink ratios (converted out of decibels into linear power ratios):

$$\left(\frac{C}{N_0}\right)_{total}^{-1} = \left(\frac{C}{N_0}\right)_{uplink}^{-1} + \left(\frac{C}{N_0}\right)_{downlink}^{-1} \tag{23}$$

This harmonic-like summation dictates that the combined link margin will always be strictly lower than the margin of the weakest individual link. For instance, if a massive rainstorm attenuates the uplink ($C/N_{up} \ll C/N_{down}$), the uplink becomes the system bottleneck, driving the total $(C/N_0)_{total}$ down asymptotically toward the uplink value. This physical coupling is a defining challenge of bent-pipe satellite engineering.

3 Advanced Analysis Models[1]

3.1 Adjacent-Satellite Interference (ASI) and IMD

In a crowded geostationary orbit (GEO) arc, satellites are spaced closely together, typically separated by only 2.0° to 3.0° of longitude. Consequently, ground station antenna beams, which have finite beamwidths, inevitably illuminate adjacent satellites, causing co-channel and cross-polar power leakage. This is known as Adjacent-Satellite Interference (ASI). Concurrently, inside the satellite transponder, multiple carrier signals are amplified by a single Traveling-Wave Tube Amplifier (TWTA). When operated near its saturation region, the amplifier's non-linear behavior causes the carriers to mix, generating unwanted Intermodulation Distortion (IMD) products that fall directly within the desired channel bandwidth.

Analyzing ASI and IMD is critical because these interference sources degrade the receiver's operating environment from a standard thermal noise channel to an interference-limited channel. Neglecting these impairments leads to overly optimistic link margins, risking severe service degradation or loss of fixed-rate closure in real-world multi-carrier environments.

The off-axis discrimination angle θ_{off} between the target satellite and the adjacent interferer is calculated using their respective ECEF position vectors. The off-axis gain $G_{\text{off}}(\theta_{\text{off}})$ in the direction of the interferer is estimated using a simplified ITU-R-style off-axis antenna envelope approximation:

$$G_{\text{off}}(\theta_{\text{off}}) = \min\left(G_{\max} - 3.0,\ 32.0 - 25\log_{10}(\theta_{\text{off}})\right) \tag{24}$$

This formula is used for parametric sensitivity analysis rather than regulatory coordination, and is valid for off-axis angles typically in the range 1°–48°. The individual carrier-to-interference (C/I) ratios for uplink, downlink, and intermodulation are combined with the thermal carrier-to-noise (C/N) ratios using the inverse-linear harmonic summation:

$$\left(\frac{C}{N+I}\right)^{-1}_{\text{total}} = \left(\frac{C}{N}\right)^{-1}_{\text{uplink}} + \left(\frac{C}{N}\right)^{-1}_{\text{downlink}} + \left(\frac{C}{I_{\text{ASI,up}}}\right)^{-1} + \left(\frac{C}{I_{\text{ASI,down}}}\right)^{-1} + \left(\frac{C}{I_{\text{IMD}}}\right)^{-1} \tag{25}$$

The combined carrier-to-noise-and-interference ratio $C/(N+I)_{\text{total}}$ acts as the key performance bottleneck, representing the true signal quality at the receiver. In dense satellite arcs with poorly isolated adjacent signals, ASI and IMD can reduce the overall link margin by several dB. In the present baseline scenario, the modeled adjacent satellites at 11.0°E and 15.0°E impose a combined interference penalty of approximately 0.37 dB, reducing the effective margin from 3.15 dB to 2.78 dB. To mitigate these effects, operators must increase ground dish diameters (narrowing the beamwidth θ_{HPBW}) or operate the satellite TWTA with sufficient back-off to suppress IMD generation.

3.2 Hardware Impairments: Non-Linearity and Phase Noise

In practice, the satellite transponder is not an ideal linear amplifier. The TWTA exhibits AM/AM compression and AM/PM conversion near saturation, while local oscillator phase noise rotates the received constellation points. These effects are collectively captured by two simplified parameters in our link budget:

[1] All topics discussed in this chapter represent optional advanced extensions developed beyond the core UZB451E term project specifications to model selected impairment cases.

- **Output Back-Off (OBO):** A deliberate reduction of the operating point below saturation to suppress non-linear distortion:

$$EIRP_{\text{active}}(\text{dBW}) = EIRP_{\text{sat}}(\text{dBW}) - OBO(\text{dB}) \tag{26}$$

- **Implementation Loss (L_{impl}):** A composite penalty (typically 1.0–2.0 dB) accounting for residual phase noise, modem non-idealities, and filter imperfections:

$$\left(\frac{E_b}{N_0}\right)_{\text{effective}} = \left(\frac{E_b}{N_0}\right)_{\text{nominal}} - L_{\text{impl}} \tag{27}$$

For the present study, an OBO of 2.5 dB and a total implementation loss of 1.2 dB are assumed, consistent with typical Ku-band TWTA operating practices [5].

3.3 Digital Baseband Performance Metrics

The satellite link must meet a strict Quality of Service, typically BER $\leq 10^{-7}$ for IP data. For coherent QPSK over AWGN, the uncoded bit error rate is:

$$P_{b,\text{uncoded}} = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) \tag{28}$$

With FEC coding (e.g., DVB-S2 LDPC+BCH), a coding gain $G_{\text{code}} \approx$ 6–8 dB shifts the operating point, enabling reliable reception at much lower raw E_b/N_0 values and pushing the link efficiency close to the Shannon limit:

$$C(\text{bps}) = B\log_2(1 + SNR) \tag{29}$$

This coding framework is the foundation for the DVB-S2 MODCOD ladder used by the ACM selector in Chapter 4.

3.4 Dynamic System Noise Temperature

In basic clear-sky link calculations, the ground station receiver system noise temperature T_{sys} is assumed to be a static constant. However, in physical links, T_{sys} fluctuates dynamically. When the ground antenna points at a low elevation angle θ, the signal must traverse a longer atmospheric path, which increases the absorbed and re-emitted background sky noise. More critically, during rain events, the lossy water molecules in the rain path absorb RF energy and simultaneously emit thermal noise, heating up the receiver's environment.

Neglecting the dynamic nature of T_{sys} during atmospheric fading events leads to a major "double-penalty" error in system design. Rain does not only attenuate the incoming carrier power (C drops), but it also simultaneously increases the system thermal noise power (N_0 spikes). This dynamic thermal-noise term is therefore useful for estimating the combined carrier attenuation, noise-temperature increase, and resulting carrier-to-noise degradation under rain-fade conditions.

The dynamic, elevation- and fade-dependent system noise temperature T_{sys} is modeled at the LNA input as:

$$T_{\text{sys}}(\theta, A_{\text{rain}}) = T_{\text{receiver}} + T_{\text{spillover}} + T_{\text{sky}}(\theta) + T_{\text{rain}}(A_{\text{rain}}) \tag{30}$$

where $T_{\text{sky}}(\theta)$ is the clear-sky noise at elevation angle θ, and T_{rain} is the rain thermal emission noise calculated using the radiative transfer equation:

$$T_{\text{rain}} = T_{\text{atm}} \left(1 - 10^{-A_{\text{rain}}/10}\right) \tag{31}$$

where $T_{\text{atm}} = 275$ Kelvin represents the physical atmospheric temperature of the lossy rain medium.

During a heavy Ku-band storm causing an attenuation of $A_{\text{rain}} = 10.0$ dB, the rain noise contribution T_{rain} spikes by approximately 247.5 Kelvin. For a high-performance ground receiver with a clear-sky $T_{\text{sys}} = 150.0$ K, this storm nearly triples the total system noise to 397.5 K. This dynamic thermal noise increase adds a 4.2 dB penalty directly to the link margin on top of the physical 10.0 dB signal attenuation, showing why dynamic noise modelling is useful for fade-margin studies.

3.5 Apparent GEO Orbit Motion

Although geostationary satellites are nominally "fixed", gravitational perturbations cause three-axis oscillations within the station-keeping box. The satellite's deviations are modeled as:

$$\Delta\lambda(t) = A_{\text{ew}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ew}}\right) \tag{32}$$

$$\Delta\phi(t) = A_{\text{ns}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ns}}\right) \tag{33}$$

$$\Delta r(t) = A_{\text{r}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}/2} + \phi_{\text{r}}\right) \tag{34}$$

These positional drifts dynamically update the slant range vector $\vec{\rho}(t)$. For the selected Rome–Ankara configuration with the modeled station-keeping amplitudes (0.06° E/W, 0.04° N/S), the computed margin variation is only approximately 0.02 dB peak-to-peak over a 48-hour window. Larger variations (often cited as ±0.2 dB) may occur for narrower antenna beams, larger pointing offsets, or more aggressive station-keeping excursions. Apparent GEO motion is not the dominant limitation for the selected link, but it provides a useful pointing robustness check. For the large uplink dish (2.4 m, narrow θ_{HPBW}), even a small drift can warrant step-tracking, whereas the consumer-grade 0.9 m receive dish remains stably locked without tracking.

4 Atmospheric Propagation and ACM[2]

This chapter presents the standards-based atmospheric attenuation models and Adaptive Coding and Modulation (ACM) mechanisms implemented in this project. **These models are additional analysis modules representing an advanced study of link availability.**

Note: The implemented attenuation workflow is based on the cited ITU-R recommendation versions. It uses a self-contained engineering implementation of the main slant-path attenuation terms; site-specific regulatory design would require exact ITU digital maps and operational coordination data. The models dynamically convert standard meteorological parameters (pressure, temperature, water vapor density, and probability of exceedance) into composite decibel attenuation penalties.

4.1 ITU-R Slant-Path Attenuation Suite

4.1.1 The Engineering Problem

Electromagnetic waves propagating between an Earth station and a GEO satellite pass through the troposphere, where atmospheric gases, cloud liquid water, scintillation, and rain introduce additional attenuation. At Ku-band, rain is especially important because the signal wavelength is comparable to hydrometeor dimensions, so scattering and absorption can reduce the received carrier power. This section estimates these atmospheric losses for the Rome–Ankara link using a standards-based engineering implementation of the relevant ITU-R propagation models.

4.1.2 Design Motivation and Rationale

To design a reliable satellite communication system, engineers must accurately predict these atmospheric fades to calculate the required system margins. The International Telecommunication Union (ITU) publishes the ITU-R propagation recommendations—empirically validated mathematical suites that allow engineers to predict slant-path attenuation at any geographic coordinate on Earth. Analyzing these atmospheric effects is an important step in high-frequency RF link design.

4.1.3 Mathematical Formulation

The total atmospheric attenuation $A_{\text{total}}(p)$ in dB exceeded for an annual time percentage p ($0.001\% \leq p \leq 5\%$) is calculated by combining gaseous absorption (A_{gaseous}), cloud attenuation (A_{cloud}), tropospheric scintillation ($A_{\text{scintillation}}$), and rain attenuation (A_{rain}) according to *ITU-R P.618-13* [6]:

$$A_{\text{total}}(p) = A_{\text{gaseous}} + \sqrt{(A_{\text{rain}}(p) + A_{\text{cloud}})^2 + A_{\text{scintillation}}(p)^2} \tag{35}$$

The point rainfall rate $R_{0.01}$ is treated as a P.837-based representative design input [7]. In the implemented scenario, $R_{0.01} = 42$ mm/h is used; for a regulatory-grade design this value should be replaced by the exact mapped value for the selected ground-station site. The frequency- and polarization-dependent specific attenuation coefficients k and

[2] All topics discussed in this chapter represent additional analysis modules developed beyond the core UZB451E term project specifications.

α are then computed from ITU-R P.838-3 [8]. The specific attenuation γ_R in dB/km is computed as:

$$\gamma_\text{R} = kR_{0.01}^{\alpha} \tag{36}$$

The effective path length through the rain cell accounts for spatial localisation:

$$L_\text{eff} = L_\text{s} \cdot r_{0.01} \cdot v_{0.01} \tag{37}$$

where $r_{0.01}$ is the horizontal path reduction factor and $v_{0.01}$ is the vertical adjustment factor. The attenuation exceeded for 0.01% of the year is $A_{0.01} = \gamma_\text{R} L_\text{eff}$. For other time percentages p:

$$A_p = A_{0.01} \left(\frac{p}{0.01}\right)^{-[0.655+0.033\ln p-0.045\ln A_{0.01}-\beta(1-p)\sin\theta]} \tag{38}$$

Gaseous attenuation uses dry air and water vapor specific zenith attenuations scaled by the slant-path elevation angle (*ITU-R P.676-12* [9]):

$$A_\text{gaseous} = \frac{\gamma_\text{o} h_\text{o} + \gamma_\text{w} h_\text{w}}{\sin\theta} \tag{39}$$

Cloud liquid water attenuation uses columnar liquid water M and specific attenuation K_l (*ITU-R P.840-8* [10]):

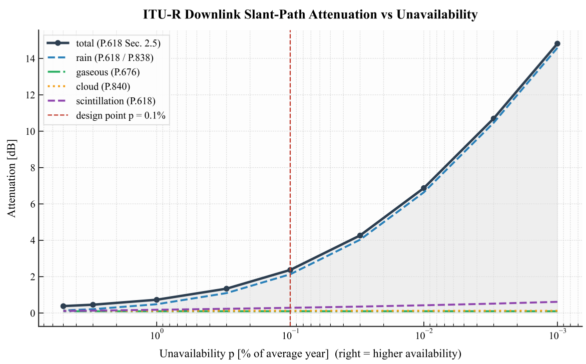
$$A_\text{cloud} = \frac{M \cdot K_\text{l}}{\sin\theta} \tag{40}$$

Scintillation fade depth is scaled by the antenna aperture-averaging factor (*ITU-R P.618-13* [6]):

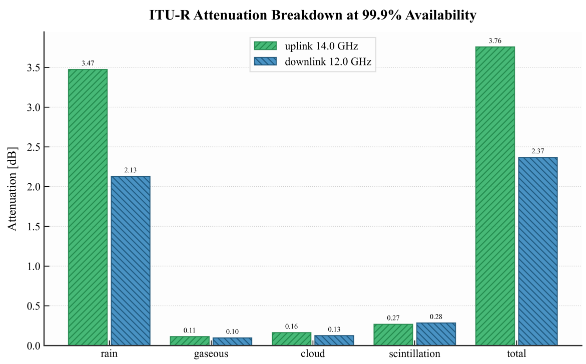
$$A_\text{scintillation}(p) = a(p)\sigma \tag{41}$$

4.1.4 Operational Impact and Results

This ITU-R suite provides the estimated fade depths that the RF link must tolerate. For our Rome-to-Ankara Ku-band link, at the 99.9% availability design point, the model predicts 3.76 dB uplink attenuation at 14 GHz and 2.37 dB downlink attenuation at 12 GHz. When these fades are combined with interference and dynamic receiver noise, the end-to-end margin falls to -1.10 dB.



(a) Slant-path attenuation vs. annual link availability.



(b) Attenuation breakdown at 99.9% availability.

Figure 4: ITU-R P.618 slant-path propagation impairments and specific contribution analysis for the Rome-Ankara link.

4.2 Monte-Carlo Rain Outage Simulation

4.2.1 The Engineering Problem

The ITU-R recommendations provide static, deterministic bounds for link availability at specific annual time percentages. However, they do not simulate the chronological, time-varying behaviour of weather—the transitions between clear-sky and storm states—nor do they provide a probability distribution of the active link margin over a continuous meteorological time series.

4.2.2 Design Motivation and Rationale

Running a Monte Carlo simulation allows system engineers to generate thousands of random meteorological states based on an illustrative weather-state probability model. This educational stochastic simulation models the statistical variance of atmospheric fades, enabling computation of the estimated probability distribution of active link margins and the estimated annual availability.

4.2.3 Mathematical Formulation

The simulation engine executes $N = 8760$ discrete trials, representing each hour of a complete year. At each time step t_i, the atmospheric weather state $S(t_i)$ is sampled from a probability mass function:

- **Clear-Sky (S_0):** Standard operating conditions (0 dB excess attenuation).
- **Light and Moderate Rain (S_1, S_2):** Small to medium fade depths drawn from a uniform attenuation profile.
- **Heavy Storms (S_3):** Rare events ($p < 0.5\%$) triggering severe attenuation (up to 15 dB in Ku-band).

For each generated hourly weather state, the computed active Carrier-to-Noise ratio and link margin are recalculated:

$$Margin(t_i) = E_b/N_0(t_i) - \text{Required } E_b/N_0 \tag{42}$$

An *outage event* is any hour where $Margin(t_i) < 0$. The overall system availability is:

$$\text{Availability (\%)} = 100 \times \left(1 - \frac{\sum \mathbf{1}(Margin(t_i) < 0)}{N}\right) \tag{43}$$

4.2.4 Operational Impact and Results

The Monte Carlo simulation outputs a probability distribution of the link margin, illustrating the estimated risk profile over the simulated year. This allows the designer to evaluate whether the selected static margin is sufficient under stochastic weather variations.

Note: This Monte Carlo model is an illustrative stochastic availability simulation based on assumed weather-state probabilities (light rain: 2.0%, moderate rain: 0.6%, heavy rain: 0.1%), not a site-calibrated meteorological time-series model derived from measured annual precipitation data. Results should therefore be interpreted as *estimated availability under the assumed stochastic weather-state model*, not as a prediction of actual annual availability.

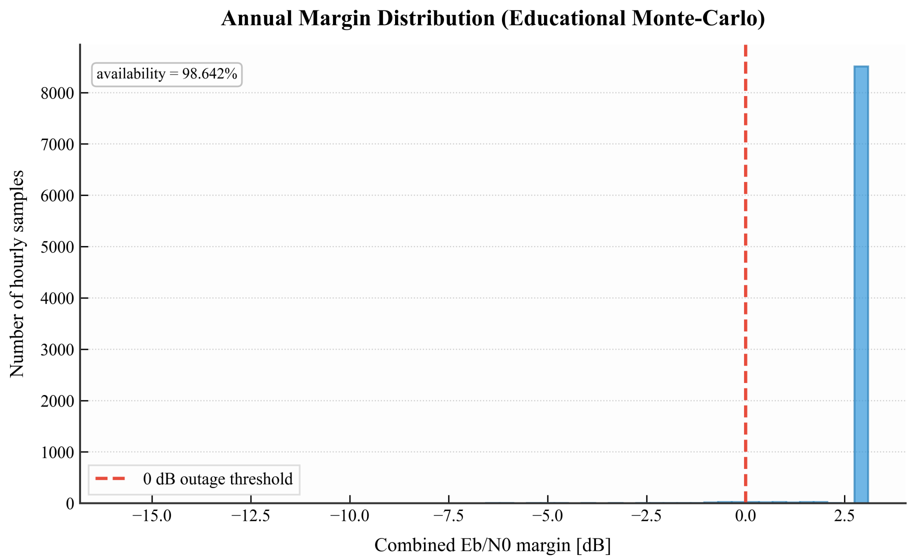


Figure 5: Probability density distribution of the dynamic link margin over 8,760 simulated hours of the meteorological year (Monte Carlo stochastic analysis).

4.3 DVB-S2 ACM (Adaptive Coding & Modulation)

4.3.1 The Engineering Problem

Traditionally, satellite links are designed for the worst-case rain fade scenario using a fixed MODCOD. Because severe storms occur for less than 1% of the year, this approach means that for over 99% of the year the link operates with a large excess margin, wasting potential channel capacity.

4.3.2 Design Motivation and Rationale

Designing with a fixed MODCOD forces a severe compromise: either accept a higher outage risk during fades or choose a conservative data rate. Adaptive Coding and Modulation (ACM) solves this by dynamically monitoring the active E_s/N_0 at the receiver and changing the modulation order and FEC code rate in real-time. This improves throughput during clear-sky conditions while reducing outage risk during fades by switching to highly robust MODCODs.

4.3.3 Mathematical Formulation

The receiver continuously monitors $(E_s/N_0)_{\text{available}}$ and dynamically selects the highest spectral efficiency η (in bits/symbol) from the DVB-S2 [11] MODCOD ladder that satisfies:

$$(E_s/N_0)_{\text{available}} \geq \text{Required } (E_s/N_0)_{\text{MODCOD}} + Margin_{\text{impl}} \tag{44}$$

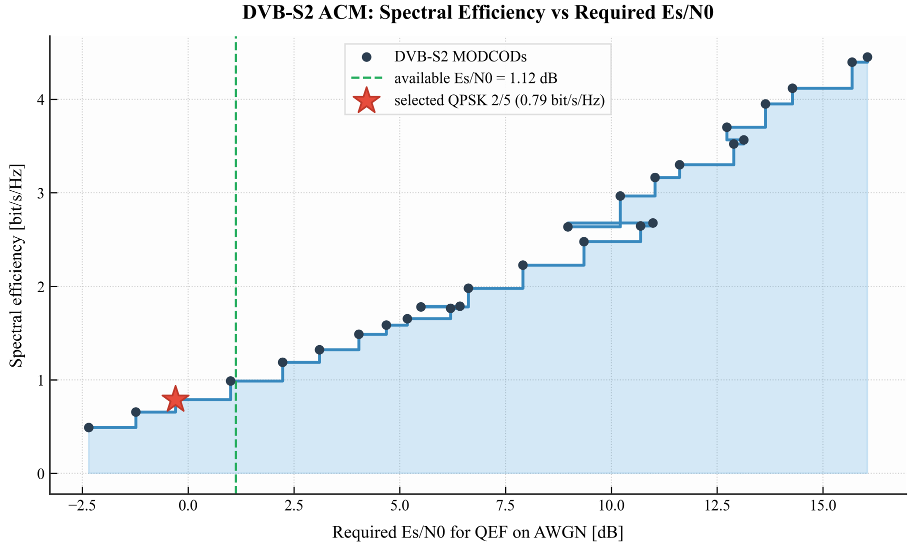


Figure 6: DVB-S2 MODCOD ladder: trade-off between spectral efficiency and required signal-to-noise ratio.

Using the usable symbol rate $R_\text{s} = B/(1 + \alpha_\text{rolloff})$, the active net throughput is:

$$Throughput(\text{Mbps}) = \eta \cdot R_\text{s} \tag{45}$$

4.3.4 Operational Impact and Results

Under clear-sky conditions, the ACM selector shifts to a higher-efficiency MODCOD than the robust rain-fade mode. The exact selected mode depends on the available E_s/N_0 and the implemented DVB-S2 threshold table. For a 36 MHz carrier with $\alpha = 0.20$ ($R_s = 30$ Msps), the available E_s/N_0 allows for QPSK 3/4 in purely thermal-noise conditions, and degrades to QPSK 2/5 under heavy rain.

Table 1 maps the specific operating points to their required thresholds, verifying the physical consistency between the link margin and the throughput claims.

Table 1: DVB-S2 ACM Consistency and Operating Point Verification

Case	Avail. E_s/N_0	Selected MODCOD	Req. E_s/N_0	Efficiency (η)	Net Throughput (ηR_s)
Clear-Sky (Thermal)	$\approx$ 5.38 dB	QPSK 3/4	4.03 dB	1.487	$\approx$ 44.62 Mbps
With ASI & IMD	$\approx$ 5.01 dB	QPSK 2/3	3.10 dB	1.322	$\approx$ 39.67 Mbps
99.9% Rain Fade	$\approx$ 1.12 dB	QPSK 2/5	-0.30 dB	0.789	$\approx$ 23.68 Mbps

Note: All points assume 1.0 dB implementation margin. Throughput is computed using $R_s = 30$ Msps.

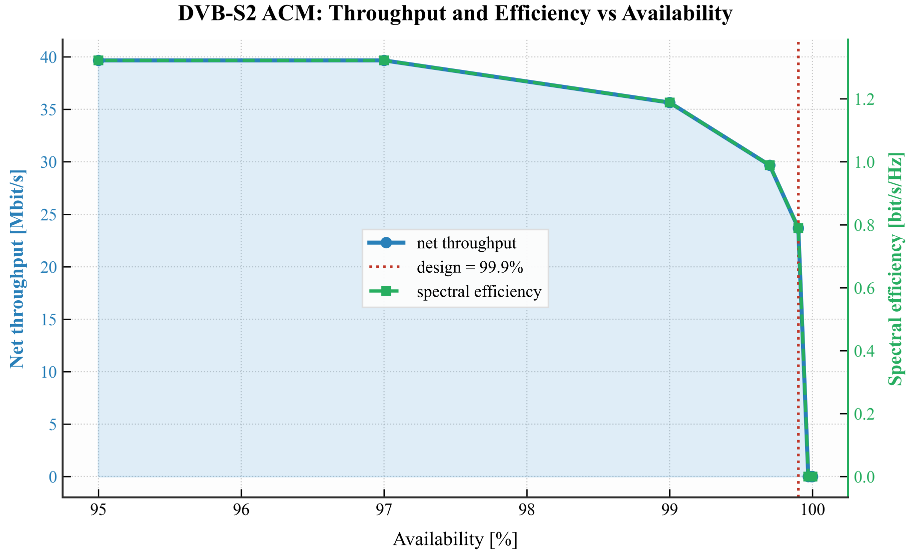


Figure 7: DVB-S2 ACM dynamic throughput and MODCOD selection over the link availability spectrum.

4.4 Uplink Power Control (ULPC)

ULPC is discussed as a mitigation concept, not as a fully implemented closed-loop control model. When heavy rain attenuates the uplink carrier, the satellite transponder input power drops below its optimal operating region. ULPC addresses this by dynamically increasing the ground station HPA transmit power:

$$P_{\text{tx,dynamic}}(\text{dBW}) = P_{\text{tx,nominal}}(\text{dBW}) + A_{\text{rain,up}}(\text{dB}) \tag{46}$$

subject to $P_{\text{tx,dynamic}} \leq P_{\text{tx,max}}$. This expression represents ideal first-order fade compensation. In practice, ULPC is constrained by HPA headroom, regulatory EIRP limits, adjacent-satellite interference budgets, transponder input back-off requirements, and the finite loop latency of the measurement-and-correction cycle. Therefore, the full recovery of the uplink attenuation through power control alone cannot always be guaranteed, and ULPC is best treated as a complementary measure alongside DVB-S2 ACM.

5 Numerical Results and Discussion

This section presents the numerical outputs computed by the structured Python analyzer and provides an engineering evaluation of the system performance.

5.1 Static Clear-Sky Link Budget Results

The clear-sky static link budget has been evaluated for the transparent Ku-band bent-pipe link between GS1 (Rome) and GS2 (Ankara) through the Eutelsat Hotbird 13G satellite at 13°E. Table 2 presents the parameter values and intermediate results computed by the software, matching the academic Pratt-style hand calculations [5].

Table 2: Static Clear-Sky Link Budget Results (Rome - Ankara).

Link Segment	**Parameter**	**Uplink**	**Downlink**	**Unit**
Geometry	Coordinates	41.90°N, 12.50°E	39.93°N, 32.86°E	–
	Satellite Longitude	13.00°E	13.00°E	–
	Slant Range (d)	$37,658.79$	$37,823.12$	km
	Elevation Angle (θ)	41.60	39.44	deg
	Azimuth Angle (Az)	179.25	209.37	deg
	Central Angle	41.91	43.85	deg
Transmitter	Frequency (f)	14.00	12.00	GHz
	Antenna Diameter	2.40	–	m
	Antenna Gain	48.86	–	dBi
	Transmitter Power	13.01 (20.0 W)	–	dBW
	EIRP	60.87	46.00 (Override)	dBW
Path Losses	Free Space Path Loss (FSPL)	206.89	205.59	dB
	Pointing Loss	0.50	0.50	dB
	Polarization Loss	0.30	0.30	dB
	Clear-Sky Atmospheric Loss	0.50	0.50	dB
	Implementation & Misc Loss	1.20	1.20	dB
	Total Link Loss (L_{total})	209.39	208.09	dB
Receiver	Antenna Diameter	–	0.90	m
	Antenna Gain	–	39.00	dBi
	Noise Temperature (T_{sys})	–	150.00	K
	Quality Factor (G/T)	3.00 (Override)	16.74	dB/K
Performance	Carrier-to-Noise Density (C/N_0)	83.08	83.25	dB-Hz
	Carrier-to-Noise Ratio (C/N)	7.51	7.69	dB
	Energy-per-Bit to Noise (E_b/N_0)	13.08	13.25	dB
	Required E_b/N_0	7.00	7.00	dB
	Link Margin	6.08	6.25	dB

The end-to-end combined carrier-to-noise density ratio is computed using inverse-linear addition:

$$\left(\frac{C}{N_0}\right)_{\text{total,linear}} = \left(10^{-83.08/10} + 10^{-83.25/10}\right)^{-1} \tag{47}$$

$$\left(\frac{C}{N_0}\right)_{\text{total,dBHz}} = 10\log_{10}\left(\left(\frac{C}{N_0}\right)_{\text{total,linear}}\right) = 80.15 \text{ dB-Hz} \tag{48}$$

yielding a combined E_b/N_0 of 10.15 dB and a combined open-air link margin of **3.15 dB**. This margin closes the link above the required digital threshold of 7.0 dB under clear-sky conditions.

5.1.1 Parametric Sensitivity and 2D Contour Map Analysis

The custom Python analyzer generates five distinct contour maps to study static system sensitivities, parameter trade-offs, and design boundaries. Each figure is analyzed below using a structured evaluation.

5.1.2 Downlink C/N0 vs. GS2 Dish Diameter and Frequency

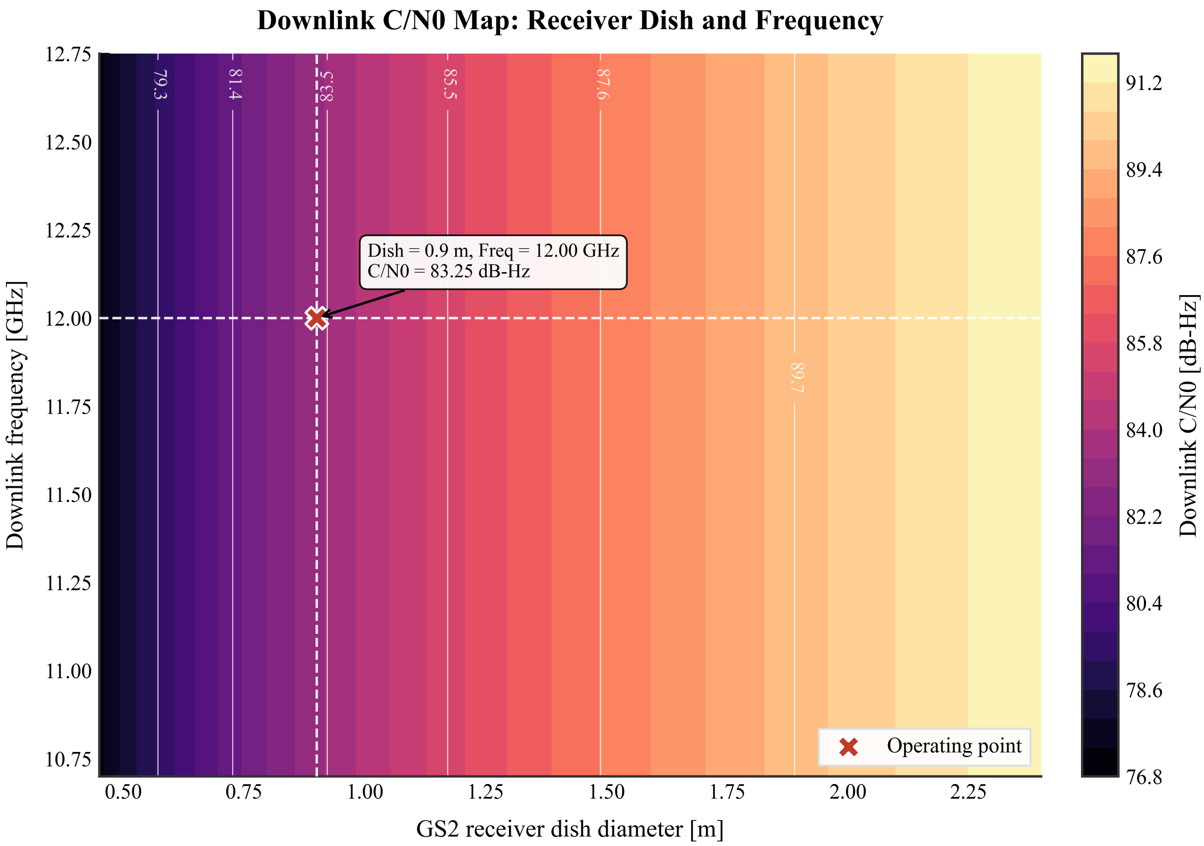


Figure 8: Downlink C/N0 vs GS2 Dish Diameter and Frequency.

- **System Setup / Objective:** Investigate how the receiving antenna aperture size (0.5 m to 2.4 m) and the downlink carrier frequency (10.7 GHz to 12.75 GHz) jointly dictate the signal power density (C/N_0) at the receiver demodulator.
- **Numerical Observations:** The C/N_0 value scales from a minimum of approximately 81.5 dB-Hz (small dish, lower end of the band) to a maximum of approximately

91.5 dB-Hz (large dish, upper end of the band). The contour lines are nearly vertical, indicating that dish diameter is the dominant variable.

- **Physical / Engineering Interpretation:** For a fixed physical aperture and fixed satellite EIRP, receive antenna gain scales as $G_{\mathrm{rx}} \propto (Df)^2$, while free-space path loss scales as FSPL $\propto f^2$. These two frequency-squared dependencies largely cancel, so the net C/N_0 is nearly independent of frequency when all other parameters are held constant. The nearly vertical contours in the figure confirm this expected cancellation. Frequency becomes a meaningful design driver only when atmospheric attenuation, antenna efficiency variation, satellite footprint EIRP variation, or band-dependent hardware assumptions are included.
- **Design Guidelines:** Ground station terminal designers utilize this map to size the minimum dish diameter required for specific satellite transponder bands, ensuring that the receiver G/T remains high enough to close the link without paying for oversized hardware.

5.1.3 Downlink Margin vs. Satellite EIRP and GS2 Tsys

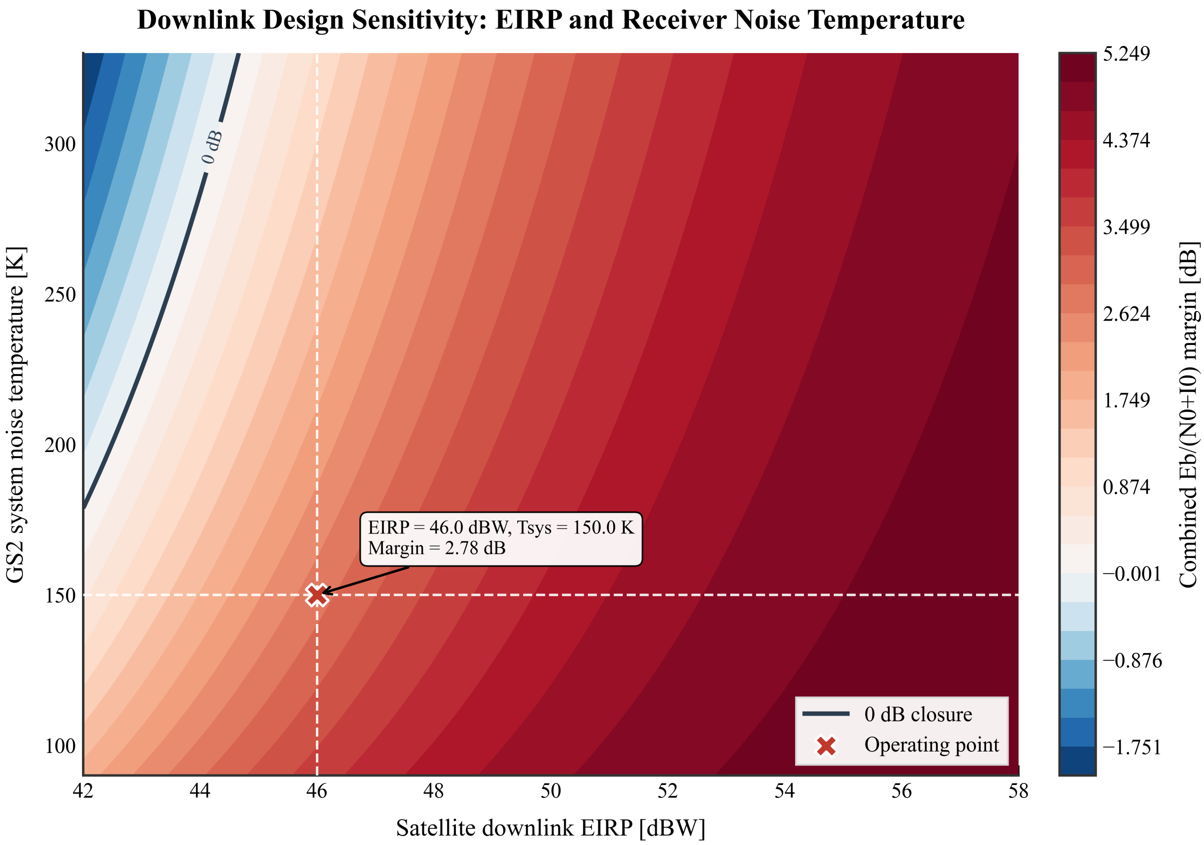


Figure 9: Combined Effective Margin vs Satellite EIRP and GS2 Tsys.

- **System Setup / Objective:** Map the combined end-to-end link margin boundary (0 dB limit) by trading off satellite transmit power (downlink EIRP, 42 dBW to 58 dBW) against the receiver noise performance (T_{sys}, 90 K to 330 K). The plotted metric is the combined Eb/(N0+I0) margin, which includes both thermal noise and the ASI/IMD interference budget active in the advanced analysis.

- **Numerical Observations:** A bold diverging-color boundary splits the parameter space at the 0 dB margin line. Any combination in the positive-margin region represents successful link closure. For instance, at a high receiver noise temperature ($T_{sys} = 300$ K), the link requires a minimum satellite EIRP of approximately 45.5 dBW to close. The operating point marker at EIRP = 46 dBW and $T_{sys} = 150$ K confirms the effective margin of approximately 2.78 dB, consistent with the advanced-impaired result summarized in Table 3.

- **Physical / Engineering Interpretation:** The combined margin varies linearly with satellite power in dB, but scales inversely with the logarithm of system noise temperature ($G/T \propto -10 \log_{10} T_{sys}$). Higher thermal noise in the receiver LNB directly degrades the signal-to-noise ratio, requiring a compensating increase in transponder power.

- **Design Guidelines:** This trade-off is critical for cost optimization. Satellite operators can decide whether to spend capital on highly powerful satellites (high EIRP) or to invest in high-performance ground terminals (low T_{sys} LNBs) to maintain a target link margin.

5.1.4 Uplink C/N vs. HPA Power and GS1 Dish Diameter

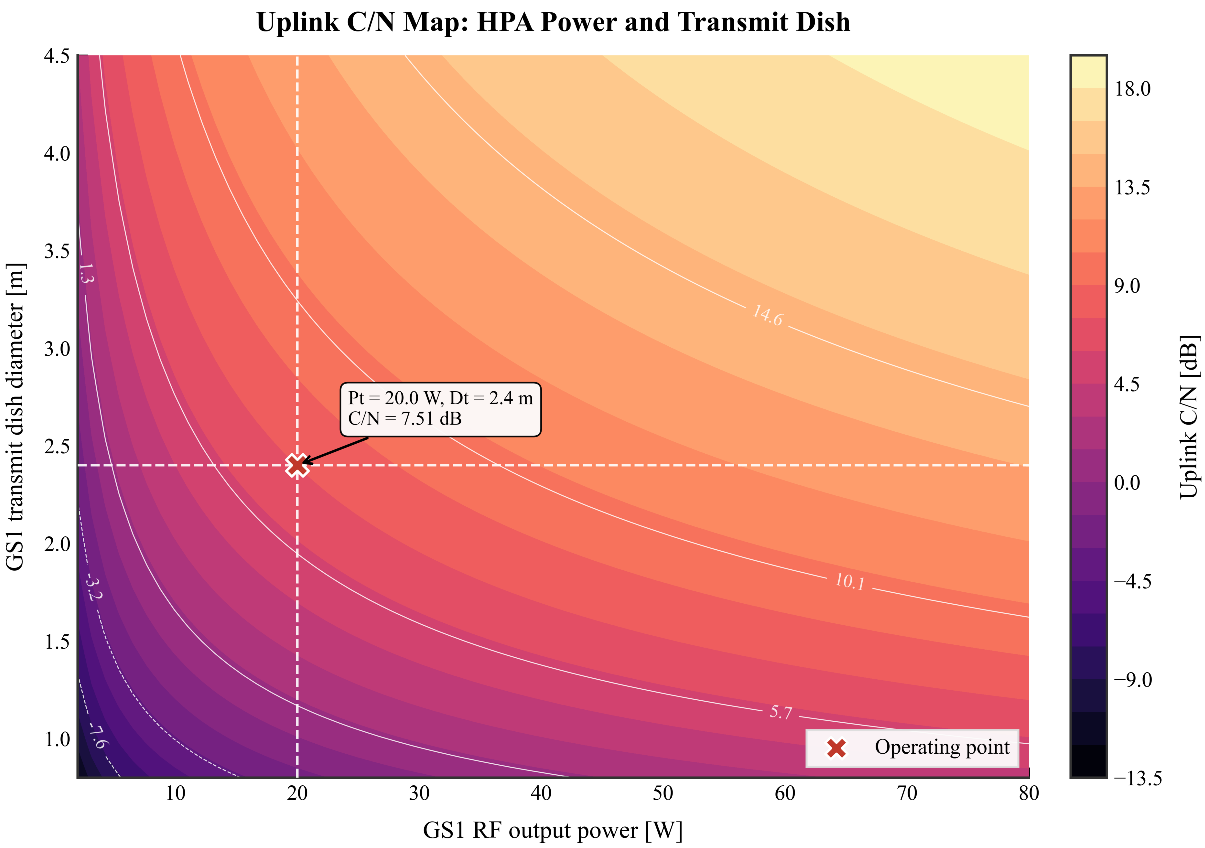


Figure 10: Uplink C/N vs HPA Power and GS1 Dish Diameter.

- **System Setup / Objective:** Evaluate the trade-off between the uplink ground station High-Power Amplifier (HPA) wattage (2 W to 80 W) and the uplink transmit antenna diameter (0.8 m to 4.5 m) to achieve a target transponder Carrier-to-Noise ratio.

- **Numerical Observations:** At the baseline operating point of 2.4 m dish and 20 W HPA, the uplink C/N is approximately 7.51 dB. Reaching a 12 dB uplink C/N target with the same 2.4 m dish would require approximately 56 W of HPA power (4.49 dB additional gain). Reducing the dish to 1.2 m introduces a further 6 dB transmit-gain penalty, requiring approximately 225 W, which lies outside the plotted range. Therefore, the contour map demonstrates that high uplink C/N targets are much more efficiently achieved by increasing antenna aperture than by increasing HPA power alone.
- **Physical / Engineering Interpretation:** Transmit gain increases by 6 dB for every doubling of the antenna diameter. This focusing gain multiplies the raw HPA wattage, allowing a major reduction in actual transmitter power to achieve the same equivalent radiated power (EIRP).
- **Design Guidelines:** HPA hardware costs scale steeply with output power. Ground terminal designers use this trade-off to minimize capital costs by choosing larger dish antennas, which also narrows the beamwidth and reduces sidelobe interference to adjacent satellites.

5.1.5 Combined Eb/N0 vs. Information Bit Rate and Bandwidth

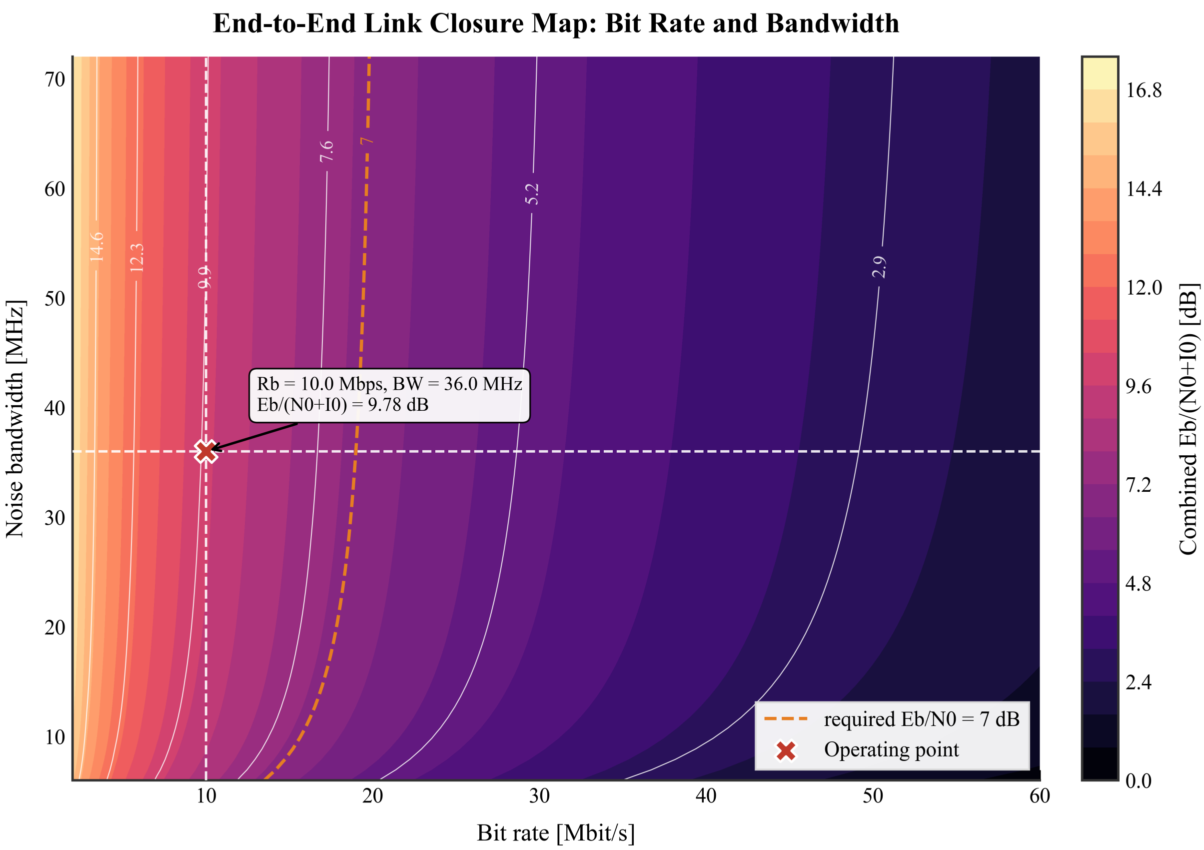


Figure 11: Effective End-to-End Eb/(N0+I0) vs Information Bit Rate and Bandwidth.

- **System Setup / Objective:** Analyze how the combined end-to-end signal energy-per-bit (E_b/N_0) is affected by changes in the information data transmission rate (2 Mbps to 60 Mbps) and the allocated noise bandwidth (6 MHz to 72 MHz).

- **Numerical Observations:** The E_b/N_0 is determined primarily by bit rate. For the clear-sky combined C/N_0 of approximately 80.15 dB-Hz, E_b/N_0 decreases from approximately 17.1 dB at 2 Mbps down to approximately 3.2 dB at 50 Mbps. At the nominal 10 Mbps operating point, the clear-sky noise-only E_b/N_0 is approximately 10.15 dB. Including adjacent-satellite interference and IMD reduces the effective value to approximately 9.78 dB. The bandwidth axis does not independently change E_b/N_0 for a fixed C/N_0 and bit rate; rather, it constrains the feasible symbol rate, occupied bandwidth, and MODCOD selection. The operating-point marker confirms the effective result consistent with the ASI/IMD-included effective result summarized in Table 3.

- **Physical / Engineering Interpretation:** Since $E_b/N_0 = C/N_0 - 10\log_{10}(R_b)$, sharing a finite carrier power C over a higher number of bits per second (R_b) directly reduces the energy allocated to each individual bit. Bandwidth determines the maximum symbol rate and occupied spectrum, but does not alter the E_b/N_0 value at fixed C/N_0 and R_b.

- **Design Guidelines:** System engineers use this chart to select the maximum data rate that can be supported without dropping the E_b/N_0 below the required FEC threshold. For data rates beyond approximately 20 Mbps, the link requires either additional satellite EIRP or a more efficient MODCOD to maintain positive margin.

5.1.6 GS2 Elevation Angle vs. Latitude and Longitude Separation

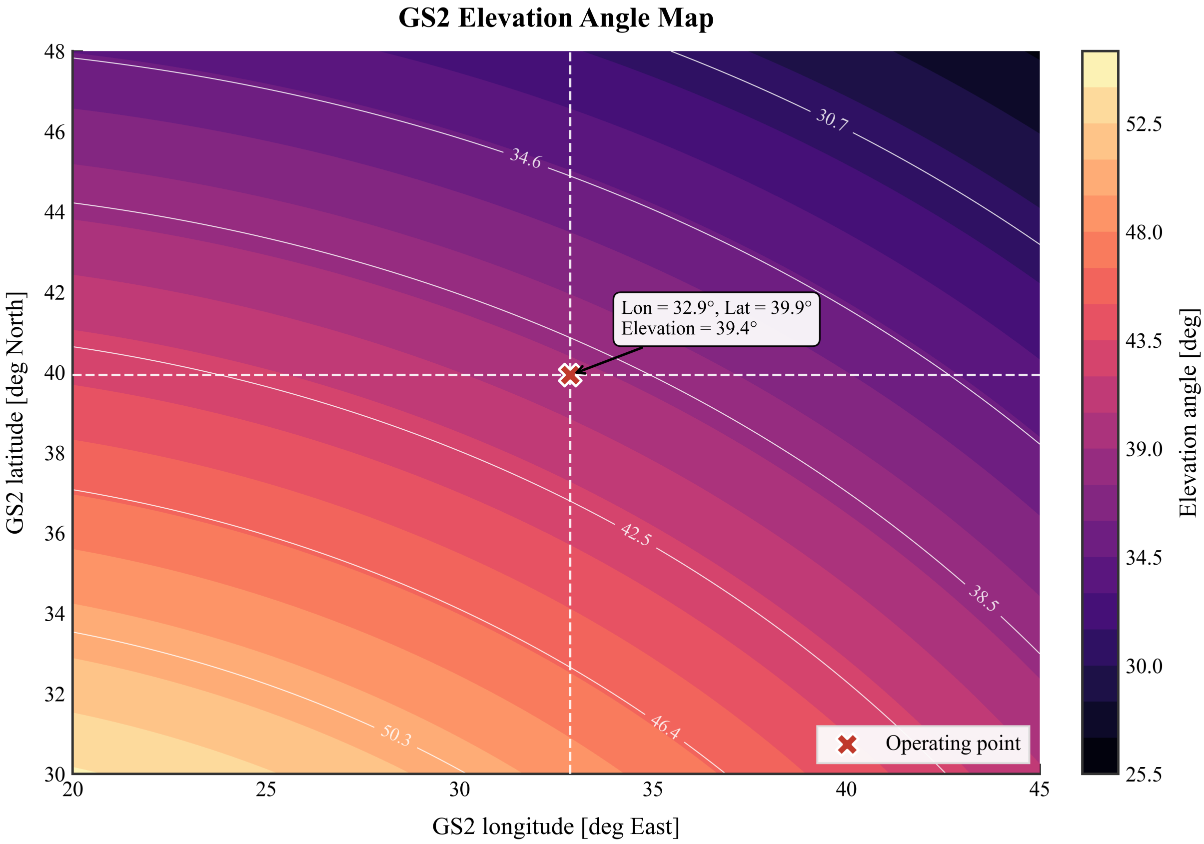


Figure 12: GS2 Elevation Angle vs Latitude and Longitude Separation.

- **System Setup / Objective:** Map the elevation pointing angle from the ground station to the geostationary satellite as a function of the station's latitude (0° to 70°N) and its longitude separation from the satellite orbit position (−60° to +60°).
- **Numerical Observations:** Concentric contour rings illustrate the elevation angle dropping rapidly as latitude and longitude separations increase, falling below the standard 10° and 5° elevation engineering limits near the edges of the map.
- **Physical / Engineering Interpretation:** Due to the curvature of the Earth, moving away from the equatorial sub-satellite point pushes the ground terminal towards the horizon, flattening the line-of-sight pointing angle and increasing the physical slant-path distance through the atmosphere.
- **Design Guidelines:** This map defines the geographic service coverage boundaries. For terminals operating in high-latitude zones (e.g., Northern Europe), system designers must allocate extra margins in the link budget to account for the severe path lengths and high tropospheric scintillation that occur at low elevation angles.

5.2 Advanced Impairment and Dynamic Analysis

This section introduces real-world atmospheric, orbital, and non-linear impairments to evaluate the system under worst-case operational conditions.

5.2.1 Interference (ASI and IMD)

- **System Setup / Objective:** Evaluate the combined link degradation when Adjacent-Satellite Interference (ASI) and Traveling-Wave Tube Amplifier (TWTA) Intermodulation Distortion (IMD) are enabled. Adjacent satellites are modeled at 11.0°E and 15.0°E longitudes relative to our target satellite at 13.0°E.

- **Numerical Observations:** Enabling ASI and IMD drives the combined Carrier-to-Noise-and-Interference ratio down, reducing the total operating margin from the clear-sky baseline of 3.15 dB down to **2.78 dB**, a penalty of approximately 0.37 dB.

- **Physical / Engineering Interpretation:** Co-channel power leakage from the adjacent satellites and non-linear intermodulation products generated within the satellite's multicarrier transponder act as extra noise sources, imposing a 0.37 dB power penalty on the link for this specific geometry. The penalty is relatively modest here because of the 2° GEO spacing and the well-sized 2.4 m uplink dish that provides strong off-axis rejection. In denser arcs or with smaller ground dishes, this penalty can be several dB.

- **Design Guidelines:** This result demonstrates that satellite operators cannot design links in isolation. Proper antenna sidelobe roll-off (to suppress ASI) and sufficient TWTA Output Back-Off (OBO $\approx$ 2.5 dB to suppress IMD) must be strictly implemented to protect the link.

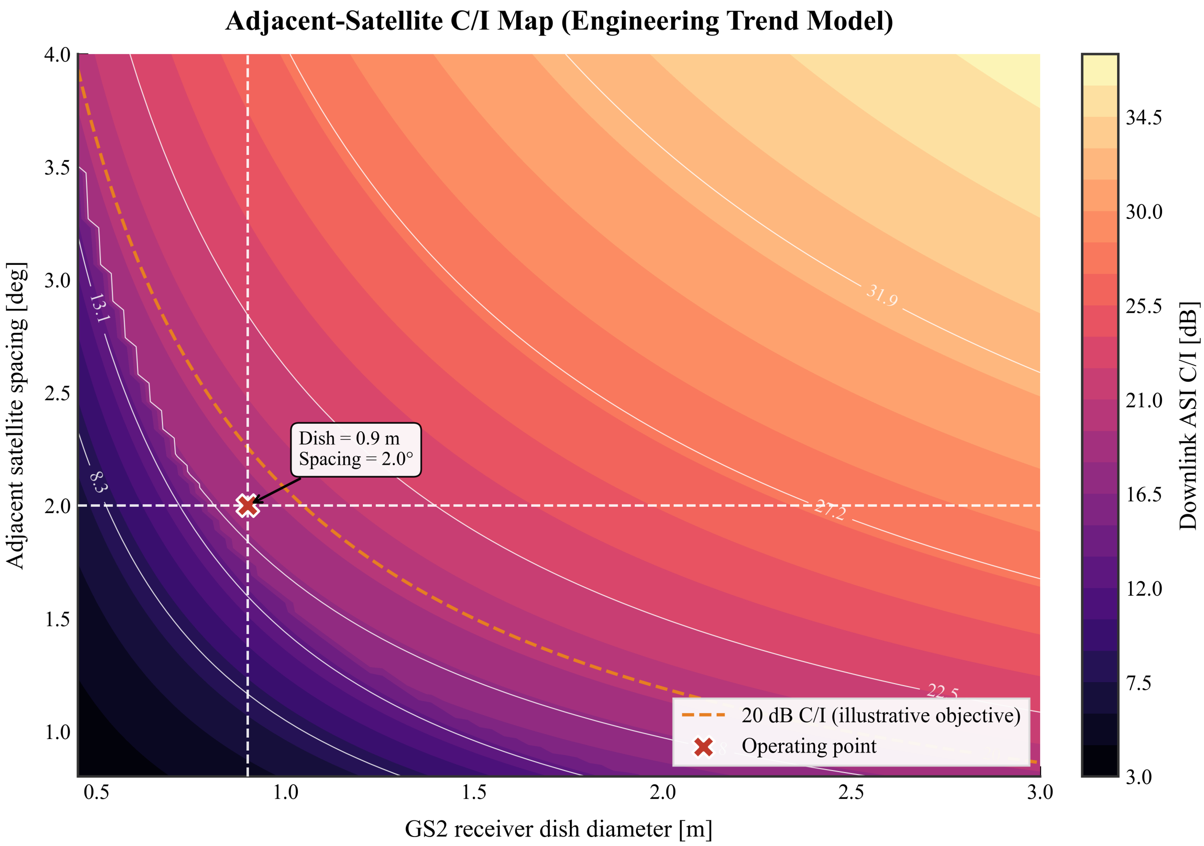


Figure 13: Downlink Adjacent Satellite Interference (ASI) C/I ratio as a function of orbital spacing and receiving antenna diameter.

5.2.2 Time-Varying Orbital Dynamics

- **System Setup / Objective:** Simulate the impact of the satellite's geostationary three-axis station-keeping drift on the path loss and link margin over a continuous 48-hour window.
- **Numerical Observations:** The sub-satellite point drift introduces a 0.02 dB peak-to-peak oscillation in the total link margin over a 48-hour period.
- **Physical / Engineering Interpretation:** Even nominally "geostationary" satellites drift within their assigned orbital station-keeping boxes due to lunar-solar gravitational perturbations and solar radiation pressure. This causes periodic variations in slant range (FSPL) and ground station elevation angles (atmospheric path length).
- **Design Guidelines:** While the 0.02 dB variation is negligible for this robust link, it validates that the software is capable of accurately tracking dynamic physical effects, which becomes critical for LEO/MEO non-geostationary orbits.

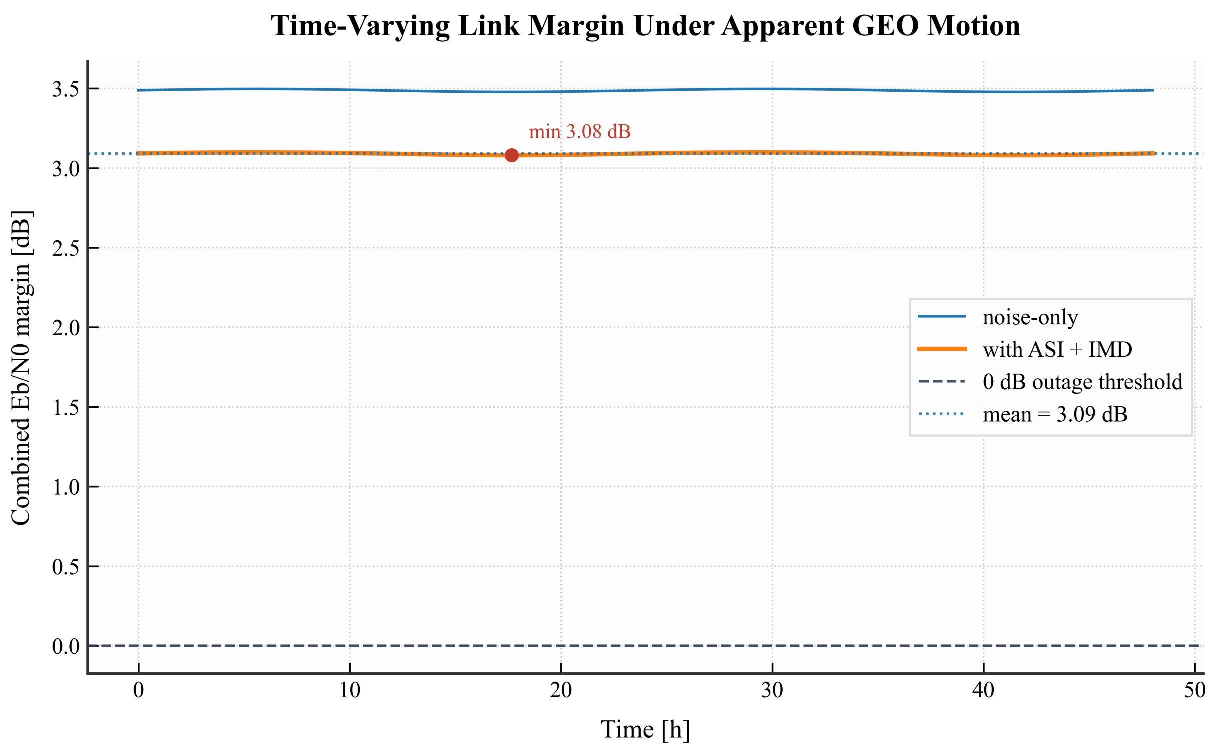


Figure 14: Time-varying combined link margin demonstrating the periodic effects of satellite apparent motion.

5.2.3 ITU-R Rain Fade and Atmospheric Impairments

- **System Setup / Objective:** Compute the excess atmospheric attenuation along the slant path under a 99.9% link availability requirement (annual exceedance probability $p = 0.1\%$) using the ITU-R propagation standards.
- **Numerical Observations:**

  - **Uplink Attenuation (14.0 GHz):** Total combined atmospheric fade of 3.76 dB.
  - **Downlink Attenuation (12.0 GHz):** Total combined atmospheric fade of 2.37 dB.

- **Physical / Engineering Interpretation:** Rain is the dominant cause of attenuation in the Ku-band, contributing to over 90% of the total fade depth. Scintillation and cloud losses represent secondary effects, while gaseous absorption is relatively static.

- **Design Guidelines:** At the 99.9% availability design point, the combined link margin falls to a negative **-1.10 dB** (with interference enabled). The baseline design does not achieve link closure at this fixed-threshold operating point. This indicates that Ku-band networks in Southern Europe and Anatolia require dynamic fade mitigation strategies such as DVB-S2 ACM, ULPC, or hardware upgrades (larger GS2 dish or higher satellite EIRP) to approach a strict 99.9% fixed-rate availability target.

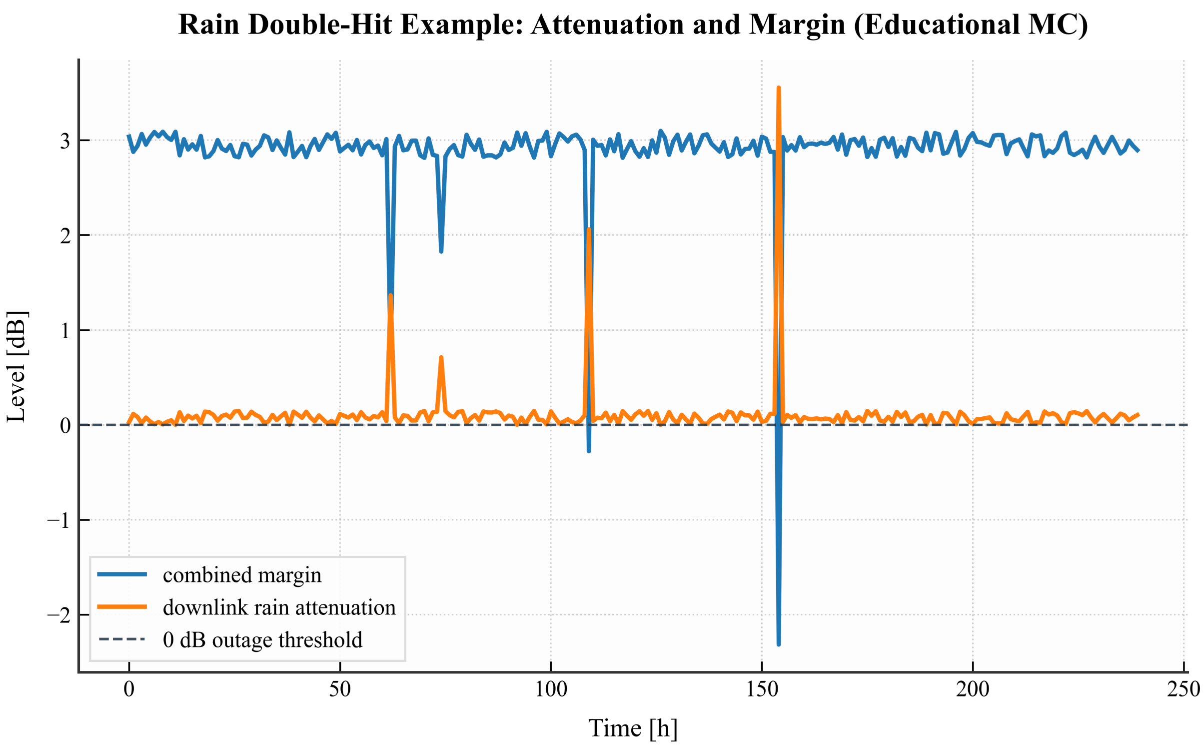


Figure 15: Time-series response of the link margin during a severe 'double-hit' weather event (simultaneous heavy rain at both the transmitter and receiver).

5.2.4 DVB-S2 ACM Throughput Performance

- **System Setup / Objective:** Evaluate how the DVB-S2 Adaptive Coding and Modulation (ACM) selector changes the transmission format during the 99.9% rain-fade case and attempts degraded-mode operation when the fixed-rate margin becomes negative.

- **Numerical Observations:** In clear-sky conditions, the link supports high-order spectral efficiencies. However, during the peak rain event, the ACM selector dynamically downshifts the MODCOD to highly robust **QPSK 2/5** (which requires a low threshold E_s/N_0 of only -0.30 dB), attempting to maintain link closure with a reduced net throughput of **23.68 Mbps**.

- **Physical / Engineering Interpretation:** By moving from QPSK 3/4 to the more robust QPSK 2/5 mode, the receiver operates at a lower required E_s/N_0 and accepts a reduced net throughput of 23.68 Mbps under the selected rain-fade case.

- **Design Guidelines:** This result shows why ACM is useful during fades: the link can trade throughput for a lower required E_s/N_0 and attempt degraded-mode operation when the fixed-rate margin is no longer positive, improving throughput during favorable link conditions while reducing outage risk during fades.

5.3 Performance Comparison and Engineering Synthesis

To fully understand the robustness of the designed spacecraft communication link, it is critical to compare the static theoretical baseline against the advanced operational model.

Assumptions and Metric Definitions

Fixed-rate metric: $E_b/N_0 = C/N_0 - 10\log_{10}(R_b)$
ACM metric: $E_s/N_0 = C/N_0 - 10\log_{10}(R_s)$
Symbol rate: $R_s = B/(1+\alpha)$
Effective impaired metric: Thermal noise and interference are combined using inverse-linear summation.
Important: Fixed-rate margin and ACM MODCOD margin are evaluated using different reference quantities (Rb vs Rs) and should not be directly interchanged.

Table 3 summarizes the key performance indicators across different simulation states.

Table 3: Performance Comparison: Static Baseline vs. Advanced Impaired Conditions.

Metric	Clear-Sky (Noise Only)	With ASI & IMD	With 99.9% Rain Fade
Link Margin	3.15 dB	2.78 dB	-1.10 dB
Available E_b/N_0 at 10 Mbps	$\approx$ 10.15 dB	$\approx$ 9.78 dB	$\approx$ 5.90 dB
Available E_s/N_0 for ACM	$\approx$ 5.38 dB	$\approx$ 5.01 dB	$\approx$ 1.12 dB
Supported MODCOD	QPSK 3/4	QPSK 2/3	QPSK 2/5
System Availability	100% (Theoretical)	100% (Clear)	99.9% (Target)
Fixed-rate 10 Mbps closure	Closed	Closed	Not closed
ACM Net Throughput	$\approx$ 44.6 Mbps	$\approx$ 39.7 Mbps	23.68 Mbps

Note: The throughput values in Table 3 represent the achievable DVB-S2 ACM net throughput for the allocated 36 MHz carrier, not the baseline 10 Mbps information-rate setting used in the fixed-rate calculations.

Interpretation of the Comparison:

1. **The Impact of Non-Linearities and Congestion:** Moving from a purely thermal-noise limited system to an interference-limited system (ASI + IMD) imposes an

immediate ≈ 0.4 dB power penalty. This demonstrates that simple static calculations may lead to optimistic margin estimates if adjacent orbital slots are congested or if the transponder is driven near saturation.

2. **The Dominance of Atmospheric Fade:** The introduction of ITU-R rain fade models drastically alters the system state. The severe attenuation in the Ku-band consumes the available 2.78 dB margin, reducing the margin to a -1.10 dB margin deficit at 99.9% availability. This indicates that the current hardware sizing is technically insufficient for a strict 99.9% SLA, requiring either more satellite EIRP, a larger GS2 dish, or relaxing the availability target.

3. **Dynamic Compensation Strategy:** When the physical margin runs out during deep fade, the DVB-S2 ACM selector downshifts to the more robust QPSK 2/5 MODCOD, allowing operation at reduced throughput under lower E_s/N_0 conditions. This reduces the net throughput to 23.68 Mbps and attempts degraded-mode operation when the fixed-rate 10 Mbps margin is no longer positive.

In synthesis, the staged analysis shows that the baseline link has positive clear-sky margin and can tolerate the selected moderate impairment cases. The static phase confirms that the link closes under clear-sky assumptions, while the advanced phase shows where the margin becomes insufficient under the 99.9% rain-fade design point. ACM can then be interpreted as a degraded-mode mitigation, but not as proof of guaranteed fixed-rate service availability. However, strict 99.9% fixed-rate link closure is not achieved by the baseline configuration (falling short by ≈ 1.10 dB) and requires additional RF margin, larger receive aperture, or higher EIRP.

6 Conclusion

This project developed and evaluated a Ku-band GEO bent-pipe link from Rome to Ankara through Hotbird 13G. The clear-sky part of the study follows the standard link-budget procedure and was implemented in a Python analysis tool. Additional modules were then used to examine interference, atmospheric attenuation, dynamic receiver noise, Monte Carlo rain states, apparent GEO motion, and DVB-S2 ACM.

The static link closes in clear-sky conditions with a combined C/N_0 of 80.15 dB-Hz, corresponding to $E_b/N_0 = 10.15$ dB at 10 Mbps and a 3.15 dB fixed-rate margin. Including the selected ASI/IMD assumptions reduces the effective margin to 2.78 dB. At the 99.9% rain-fade design point, the ITU-R-based attenuation case reduces the fixed-rate margin to -1.10 dB, so the baseline hardware does not meet strict 99.9% fixed-rate closure without additional margin.

The ACM analysis shows that the link can trade throughput for a lower required E_s/N_0. For the 36 MHz carrier and 0.20 roll-off factor, the selector moves from QPSK 3/4 in clear-sky conditions to QPSK 2/5 in the rain-fade case, giving an estimated net throughput of 23.68 Mbps if the selected MODCOD still closes. This should be interpreted as a degraded-mode mitigation rather than proof of guaranteed service availability.

Based on these results, the most direct improvements are increasing the GS2 receive dish diameter, increasing available downlink EIRP, adding uplink power-control headroom, or reducing the fixed-rate data-rate requirement during severe fades. A more detailed design would also use exact site-specific P.837 rainfall data, measured receiver noise temperature, and real transponder coordination data.

Source Code Availability:
The complete custom Python simulation framework, including the Monte-Carlo and ITU-R propagation models developed for this project, is open-source and publicly available at: https://github.com/ayberkdt/link_budget_analysis.

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